

PRODUCT REQUIREMENTS DOCUMENT

Inventory Barcode Generator with Asset Tracking

Functional Prototype — v1.0

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Target Segment: 100–5,000 SKU Small Businesses	Price Point: \$39/month flat
Personas: Jayesh, Karthik, Hatem, Sarah, Priya	Format: Google Sheets Native + PWA

This PRD synthesizes problem statement analysis, real user personas, empathy mapping, competitive analysis, and customer journey mapping to define a functional prototype for the Inventory Barcode Generator with Asset Tracking platform.

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1. EXECUTIVE SUMMARY

The Inventory Barcode Generator with Asset Tracking is a SaaS platform designed for small businesses (100–5,000 SKUs) transitioning from manual spreadsheet-based inventory tracking to their first barcode system. The platform eliminates the need for expensive hardware (\$300+ dedicated scanners) and complex ERP implementations by leveraging smartphone cameras, Google Sheets as the source of truth, and a progressive web app (PWA) for offline-capable scanning.

Core Value Proposition

Reduce inventory management time from 4+ hours to 30 minutes, eliminate stockouts and lost items, and maintain the existing spreadsheet workflow while adding barcode superpowers — all at \$39/month with unlimited users and items.

Three-Step Setup

Step 1: Upload	Step 2: Generate	Step 3: Print & Scan
Import existing inventory spreadsheet (CSV/XLSX). No data re-entry required. System auto-detects columns: SKU, Name, Location, Quantity.	Auto-generate unique Code 128 barcodes with customizable prefixes (e.g., SS-001). Sequential numbering patterns. Bulk generation for 2,000+ SKUs in seconds.	Download PDF label templates for any standard printer. Cut and tape, or use Avery labels. Scan with smartphone PWA — no dedicated scanner needed.

Target Outcomes

- Reduce inventory count time from 4+ hours to 30 minutes
- Find any item in seconds via smartphone scan (vs. 20+ minutes of hunting)
- Track who last handled expensive equipment with timestamped audit trail
- Maintain accurate stock levels without manual counting
- Prevent stockouts and lost items costing hundreds in lost sales and replacement costs

2. PROBLEM STATEMENT & MARKET CONTEXT

2.1 The Spreadsheet Trap

Small businesses with 100–5,000 SKUs currently track inventory in spreadsheets but lose hours weekly to manual counts and cannot locate items quickly. The gap between what the spreadsheet says and where the item actually is keeps widening between monthly counts.

2.2 Pain Points by Frequency

Pain Point	Frequency	Business Impact	Persona Most Affected
Manual inventory counts	Weekly (4–6 hours)	Labor cost, weekend disruption, family time lost	Sarah (Retail), Karthik (Mfg)
Cannot locate items quickly	Daily (3–5 incidents)	Customer wait, missed sales, employee frustration	Sarah (Retail), Jayesh (Retail)
Stockouts from inaccurate counts	Weekly (2–3 SKUs)	Lost sales, emergency reorders, reputation damage	Jayesh (Retail), Aiden (E-com)
No audit trail for equipment	Monthly (1–2 disputes)	Unaccounted losses, blame games, insurance claims	Karthik (Mfg), Hatem (Construction)
Spreadsheet version chaos	Daily (multiple edits)	Decisions on stale data, duplicate orders, waste	All personas
Compliance/documentation gaps	Per audit cycle	Fines, contract loss, license risk	Dr. Priya (Medical), Hatem (Construction)

2.3 Market Gap

The market has two extremes: (1) Free online barcode generators with no scanning, sync, or tracking, and (2) Enterprise ERP systems (\$999+/month) requiring 3+ week implementation and dedicated IT. There is no solution that sits in the just right zone for first-time barcode adopters who need Google Sheets integration, phone-based scanning, and flat pricing with unlimited users.

3. TARGET USERS & PERSONAS

3.1 Primary User Segments

Segment	Vertical	SKU Range	Tech Comfort	Core Trigger
Jayesh Suchde	Pop-Culture Retail	500–2,000	High	Movie-release stockouts
Karthik Synerg	Manufacturing	1,000–3,000	Medium	MD demands variance fix
Hatem Al Shahed	Construction/Oilfield	500–2,000	Medium	Client audit crisis
Sarah Mitchell	Independent Retail	800	Medium	Sunday count exhaustion
Dr. Priya Sharma	Medical Practice	600	Medium	State inspection failure

3.2 Persona Deep Dive: Jayesh Suchde (Primary)

Role: Senior VP, Sourcing at The Souled Store (India's largest pop-culture merchandise brand)
Context: 500+ SKUs, 15 retail stores, 50+ vendors across India/China/Bangladesh
Current Stack: Excel master sheet (2,000 rows, 4 editors), Shopify for online, WhatsApp for store communication
Core Pain: Movie tie-in merchandise has 6–8 week selling windows; dead stock is a margin killer. System says 40 units, floor says 12. Spends Sunday evenings reconciling.

Empathy Map — Jayesh

Says: My assortment team needs real-time stock levels before they plan the next Marvel drop. / Show me the margin impact, not the feature list.
Thinks: If we miss a movie release window, that entire collection becomes dead stock. / My Excel master sheet has 2,000 rows and 4 people edit it — it is a ticking time bomb.
Does: Spends Sunday evenings reconciling warehouse counts with store sales data. Calls store managers every morning for what is low updates. Manually creates POs based on gut feel when Excel is outdated.
Feels: Anxious during movie release weeks. Frustrated when a hot SKU shows available online but is physically missing. Vindicated when inventory numbers match physical counts (rare).

3.3 Persona Deep Dive: Sarah Mitchell (Secondary)

Role: Owner, Mitchell's Home Goods (Independent home décor/gift shop, Austin, TX)
Context: 800 SKUs, 3 part-time employees, online (Shopify) + physical store
Current Stack: Handwritten tags + Excel (master list updated monthly), Shopify inventory
Core Pain: Spends every Sunday 6 AM–12 PM with clipboard. Sells items online that are actually out of stock. Tried a POS system once — \$200/month, 3-week setup, staff revolted, canceled month 2.

Empathy Map — Sarah

Says: I know I have it somewhere — let me check the back room. / I cannot afford another \$200/month thing that my staff ignores.
Thinks: If I spend \$39/month and it does not work, that is \$468/year I could have spent on inventory. / My competitor has those handheld scanners — \$300 each. For three employees? No.
Does: Spends every Sunday 6 AM–12 PM walking the store with clipboard and pen. Hand-writes price tags. Checks online orders against physical stock by walking to the item.
Feels: Tired on Sunday afternoons. Embarrassed when a customer waits while she hunts for an item. Skeptical of tech solutions — They all promise, none deliver for a store my size. Hopeful — There has to be a simpler way, right?

4. PRODUCT VISION & VALUE PROPOSITION

4.1 Vision Statement

To become the default first barcode system for small businesses worldwide — the bridge between spreadsheet chaos and operational clarity, without requiring data migration, hardware investment, or IT support.

4.2 Value Proposition Canvas

Customer Jobs	Pains	Gains
Track inventory accurately across locations Find items quickly when needed Maintain compliance/audit readiness Scale operations without proportional headcount	Manual counts consume weekends Stockouts damage revenue and reputation No audit trail for expensive items Spreadsheet version chaos Staff resistance to complex systems	Real-time visibility from anywhere Reduced labor hours Accurate stock levels Audit-ready documentation Staff adoption without training overhead
Products & Services	Pain Relievers	Gain Creators
Bulk Code 128 barcode generation Printable PDF label templates PWA mobile scanner (offline) Google Sheets real-time sync Check-in/check-out workflows Movement history & audit trail	Upload existing spreadsheet — no migration Any standard printer — no hardware Phone camera — no \$300 scanner Google Sheets — no new system to learn Offline mode — works in any warehouse Flat pricing — no per-user fees	30-minute counts vs. 4+ hours Find items in 3 seconds vs. 20+ minutes Timestamped who last handled tracking Auto low-stock alerts Multi-location transfer logs Investor-ready inventory reports

4.3 Competitive Positioning

Against Free Tools (Online Barcode Generators): We offer scanning, sync, tracking, and workflows — not just static barcode images.

Against inFlow/Sortly (\$49–\$99/month + per-user fees): We offer Google Sheets as source of truth (no migration), flat pricing (unlimited users), and deeper offline PWA capabilities.

Against Fishbowl/Cin7 (\$229–\$349/month): We offer sub-hour setup vs. 3+ week implementation, no IT required, and no per-user fees that scale with team size.

Against ERP (NetSuite/\$999+): We offer immediate activation, no consultant fees, and no organizational change management — built for businesses that just need to stop losing items.

5. FUNCTIONAL REQUIREMENTS

Priority: P0 = Must have for MVP, P1 = Should have for v1.1, P2 = Nice to have for v1.2

5.1 Barcode Generation Module (P0)

ID	Requirement	Acceptance Criteria	Persona
BG-001	Upload inventory spreadsheet (CSV/XLSX)	Auto-detect columns: SKU, Name, Description, Location, Quantity, Category. Support files up to 10MB (50,000 rows). Handle missing/duplicate SKUs with user confirmation dialog.	All
BG-002	Generate Code 128 barcodes in bulk	Generate unique barcodes for all uploaded SKUs with customizable prefix (e.g., SS-, MHG-, DVM-). Sequential numbering with configurable start number and padding (e.g., SS-0001). Support 2,000+ SKUs generated in under 30 seconds.	Jayesh, Sarah
BG-003	Barcode preview and validation	Show preview of first 10 barcodes before bulk generation. Validate scannability using industry-standard Code 128 validation. Alert if barcode contains invalid characters or exceeds length limits.	All
BG-004	Export barcode data to Google Sheets	Auto-create or update Google Sheet with columns: SKU, Barcode Number, Product Name, Location, Quantity, Last Scan Date, Current Location. Append barcode columns to existing sheet without overwriting user data.	All
BG-005	Regenerate barcodes for new items	Upload updated spreadsheet with new SKUs. System detects existing vs. new items. Generates barcodes only for new items while preserving existing barcode assignments.	Jayesh, Karthik

5.2 Label Printing Module (P0)

ID	Requirement	Acceptance Criteria	Persona
LP-001	Printable PDF label templates	Generate PDF with multiple label layouts: standard paper (cut-and-tape), Avery 5160/8160 (30 per sheet), Avery 5163 (10 per sheet). Support custom dimensions input (width x height in mm/inches).	Sarah, Jayesh
LP-002	Label content customization	User selects which fields to print on label: Barcode (required), Product Name, Location, SKU, Price, Custom Field. Font size auto-scales to fit label dimensions. Preview before download.	All
LP-003	Print on any standard printer	PDF optimized for laser and inkjet printers (300 DPI minimum). No special label printer required. Tested on HP, Canon, Epson home/office printers.	Sarah, Dr. Priya
LP-004	Bulk label printing by filter	Filter by location, category, or SKU range. Print labels for subset of inventory (e.g., only Aisle 3 items, only new arrivals this month).	Jayesh, Karthik
LP-005	Barcode reference sheet	Generate single-page reference sheet with all barcodes and product names for quick desk lookup. Useful for front-desk staff who need to scan without walking to shelf.	Sarah, Dr. Priya

5.3 PWA Mobile Scanner Module (P0)

ID	Requirement	Acceptance Criteria	Persona
SC-001	Camera-based barcode scanning	Use device camera to scan Code 128 barcodes. Support continuous scanning mode (scan multiple items without tapping between each). Auto-focus and auto-exposure handling for warehouse lighting conditions. Scan success rate > 95% in typical indoor lighting.	All
SC-002	Instant item lookup	On successful scan, display: Product Name, Current Location, Quantity, Last Scan Date, Last Scanned By. Lookup time < 2 seconds on 4G connection. Show red alert if item not found in database.	Sarah, Jayesh
SC-003	Check-in/check-out workflow	Scan item → select action: Check In (add to stock), Check Out (remove from stock), Transfer (move to new location). Require optional note/reason field. Auto-timestamp and log user identity.	Karthik, Hatem
SC-004	Location transfer with timestamp	Scan item → select Transfer → choose destination location from dropdown or type new location. Record: From Location, To Location, Transfer Time, Transferred By. Update Google Sheets in real-time or queue for sync.	Jayesh, Hatem
SC-005	Offline mode with auto-sync	Store scan actions locally when no network (IndexedDB). Queue up to 500 actions offline. Auto-sync when connection restored. Show sync status indicator: pending count, last sync time, sync errors.	Hatem, Karthik
SC-006	Multi-user support (no login friction)	Generate unique PWA link per team/location. Users enter name/ID on first open (no password required). All scans attributed to that name until changed. Admin can view scan history by user.	Sarah, Dr. Priya

5.4 Google Sheets Integration Module (P0)

ID	Requirement	Acceptance Criteria	Persona
GS-001	One-click Google Sheets connect	OAuth 2.0 authentication with Google. User selects existing spreadsheet or creates new. Grant read/write permissions for specific sheet only (not full Drive access). Connection setup < 2 minutes.	All
GS-002	Auto-append barcode columns	System adds columns to existing sheet without overwriting: Barcode Number, Last Scan Date, Current Location, Last Scanned By, Quantity (synced). Preserves all user-existing columns and formulas.	Jayesh, Sarah
GS-003	Real-time bidirectional sync	Scan in PWA → update Sheet within 5 seconds. Edit quantity in Sheet → reflect in PWA within 30 seconds (polling). Handle conflicts: last-write-wins with timestamp log. Show sync status in both web app and PWA.	All
GS-004	Google Drive backup	Auto-save version history in Google Drive (native Sheets feature). User can view/edit from any device with Google account. No separate database login or password to remember.	Sarah, Dr. Priya
GS-005	Conditional formatting for alerts	Auto-apply conditional formatting in Sheets: Red background when Quantity < Reorder Point. Yellow when Quantity < Reorder Point x 1.5. Green when Quantity >= Reorder Point x 2. User can customize thresholds.	Jayesh, Dr. Priya

5.5 Asset Tracking & Alerts Module (P1)

ID	Requirement	Acceptance Criteria	Persona
AT-001	Movement history log	Record every scan action: Item, Action (In/Out/Transfer), From Location, To Location, Timestamp, User, Note. View full history in web app with filter by date range, item, user, or location. Export to CSV.	Karthik, Hatem
AT-002	Low-stock alerts	User sets reorder point per SKU. Alert triggered when Quantity <= Reorder Point. Alert methods: email, in-app notification, highlighted row in Google Sheets. Alert includes: SKU, Product Name, Current Quantity, Reorder Point, Suggested Order Quantity.	Jayesh, Sarah
AT-003	Multi-location inventory	Support unlimited locations (Warehouse, Store A, Store B, Site 1, etc.). View inventory by location in web app and Sheets. Transfer items between locations with full audit trail. Location-specific reorder points.	Jayesh, Hatem
AT-004	Basic reporting dashboard	Web app dashboard shows: Total SKUs, Total Items, Items by Location, Low Stock Count, Recent Transfers (last 7 days), Top Movers (most scanned items). Export any report to PDF or CSV.	Jayesh, Karthik

AT-005	Expiration date tracking (Medical/Perishable)	Optional column: Expiration Date. Alert 30/60/90 days before expiration. Red highlight in Sheets when expired. Movement history shows lot numbers for traceability. Required for Dr. Priya medical compliance use case.	Dr. Priya
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6. NON-FUNCTIONAL REQUIREMENTS

6.1 Performance Requirements

Requirement	Target	Measurement
Barcode generation speed	2,000 SKUs in < 30 seconds	Server-side processing time from upload to downloadable barcodes
PDF label generation	500 labels in < 10 seconds	Time from click to PDF download ready
PWA scan lookup	< 2 seconds on 4G	Time from successful scan to item details displayed
Google Sheets sync latency	< 5 seconds (online)	Time from scan action to Sheet update visible
Offline queue capacity	500 actions minimum	Number of scan actions storable without network
Page load time	< 3 seconds (web app)	Time to interactive on standard broadband
Concurrent users	100+ per account	Simultaneous PWA users scanning without degradation

6.2 Security Requirements

Requirement	Implementation	Rationale
OAuth 2.0 for Google Sheets	Standard Google OAuth with scoped permissions (spreadsheets.readonly, spreadsheets, drive.file). No full Drive access requested.	Minimize permission scope to reduce user trust barrier. Only access user-explicitly-selected spreadsheet.
PWA link security	Unique UUID-based PWA URLs per account. Optional PIN protection for sensitive environments (medical, controlled substances).	Balance accessibility (no login friction) with security for high-risk verticals.
Data encryption	TLS 1.3 for all data in transit. AES-256 for data at rest (scan logs, user metadata).	Industry standard. Required for medical and financial compliance contexts.
No PII storage	Store only: account email, team member names (user-provided), scan actions. No credit card data (handled by Stripe). No customer data from Sheets.	GDPR/CCPA compliance. Minimize data liability.
Audit log retention	Retain scan action logs for 2 years. User can export or delete (GDPR right to erasure).	Medical compliance (Dr. Priya) and manufacturing traceability (Karthik) require multi-year records.

6.3 Reliability & Availability

Requirement	Target	Implementation
Uptime SLA	99.5% monthly uptime	Cloud hosting (AWS/GCP) with auto-scaling, health checks, and automated failover.
Offline functionality	100% offline scan capability	PWA Service Worker caches app shell. IndexedDB stores scan queue. Background sync when online.
Data backup	Real-time (Google Sheets native)	Google Drive version history provides automatic backup. User can restore any prior version from Google Sheets interface.
Disaster recovery	RPO < 1 hour, RTO < 4 hours	Database snapshots every hour. Multi-region deployment for critical path services.

7. USER FLOWS & WIREFRAME LOGIC

7.1 Primary Flow: First-Time Setup (Sarah Mitchell Use Case)

Flow Steps:

1. **Landing Page:** Sarah sees headline: "Turn your spreadsheet into a scannable warehouse in 20 minutes." CTA: "Start Free Trial — No credit card."
2. **Sign Up:** Email + password, or Google Sign-In (one click). Account created.
3. **Upload Spreadsheet:** Drag-and-drop CSV/XLSX. System auto-detects columns. Sarah confirms mapping: Column A = SKU, Column B = Product Name, Column C = Location, Column D = Quantity.
4. **Preview & Validate:** System shows first 10 rows with detected columns. Sarah spots one error: "Location" column detected as "Category." She drags to correct.
5. **Generate Barcodes:** Sarah enters prefix "MHG-" and start number "0001". Clicks "Generate." 800 barcodes created in 12 seconds. Preview grid shows first 20.
6. **Connect Google Sheets:** One-click OAuth. Sarah selects her existing "Inventory Master" Sheet. System appends columns: Barcode Number, Last Scan Date, Current Location. All 800 rows populated.
7. **Print Labels:** Sarah selects "Avery 5160 — 30 per sheet." Clicks "Download PDF." Opens in browser, prints on home office HP printer.
8. **Get PWA Link:** System generates unique link: `app.inventorybarcode.com/pwa/abc123`. Sarah texts to employee Emma.
9. **First Scan:** Emma opens link, enters name, scans first item. Screen shows: "Ceramic Planter Sage Green — Aisle 2, Shelf B — 3 units — Last scanned: Never." Emma taps "Confirm Location."
10. **Confirmation:** Sarah checks Google Sheets. Row updated: Last Scan Date = now, Current Location = Aisle 2, Shelf B. Setup complete. Time elapsed: 18 minutes.

Wireframe Logic:

- Each step is a single screen (no scrolling required on desktop).
- Progress indicator: Step X of 5 (collapses Upload/Preview into one step).
- No "Save" buttons — auto-save at each step with undo option.
- Help tooltip on every screen: "Stuck? Watch 2-min video" or "Chat with us."

7.2 Secondary Flow: Crisis Audit (Hatem Al Shahed Use Case)

Flow Steps:

1. **Emergency Sign-Up:** Hatem discovers tool via Google search at 11 PM. Signs up with Google account (no form filling).
2. **Rapid Upload:** Uploads supplier invoice list (600 SKUs, Excel). Auto-detect works perfectly — no corrections needed.
3. **Dual-Language Labels:** Hatem selects "English + Arabic" label format. System generates labels with English SKU + Arabic description. Prints on office printer.
4. **Offline PWA Test:** Dispatches labels to JAFZA warehouse and 3 remote sites. Site supervisor in Empty Quarter downloads PWA, tests scan with airplane mode ON — works.
5. **Lot Number Tracking:** Hatem enables "Lot Number" column in Sheet. Scan now reveals: Supplier, PO Date, Customs Entry Number.
6. **Audit Day:** Client asks for Lot #GST-2024-044 traceability. Hatem opens Google Sheets, filters by Lot Number, shows full chain: Received JAFZA March 10, Transferred Site B March 15, Scanned by Ahmed March 15 14:32.
7. **Audit Pass:** Zero discrepancies. Contract renewal approved. Hatem upgrades from trial to annual plan.

Wireframe Logic:

- Crisis flow prioritizes speed: every screen has "Skip for now" option with smart defaults.
- Offline test is a dedicated onboarding step: "Test your connection" with airplane mode toggle simulation.
- Audit view is a one-click report: "Generate Audit Report" button on dashboard.
- Annual plan upsell appears after first successful audit with ROI calculator: "You just saved \$5,000 in liquidated damages. Annual plan = \$468."

8. DATA MODEL & ARCHITECTURE

8.1 High-Level Architecture

Frontend Layer:

- Web Application (React/Vue): Barcode generation, label printing, dashboard, Google Sheets OAuth
- PWA (React/Vue + Service Worker): Camera scanning, offline queue, sync management

Backend Layer:

- API Server (Node.js/Python): RESTful API for CRUD operations, barcode generation, PDF generation
- Google Sheets Sync Service: Webhook/polling-based bidirectional sync
- Notification Service: Email alerts, in-app notifications

Data Layer:

- Primary Database (PostgreSQL): User accounts, scan logs, movement history, metadata
- Cache (Redis): Session management, rate limiting, sync state
- Object Storage (S3/Cloud Storage): Generated barcode images, PDF labels, reference sheets

External Integrations:

- Google Sheets API (OAuth 2.0): Source of truth for inventory data
- Google Drive API: Backup and version history
- Stripe: Subscription billing
- Email Service (SendGrid/AWS SES): Alerts and notifications

8.2 Core Data Entities

Entity	Key Attributes	Source of Truth	Notes
Account	id, email, plan_type, created_at, google_sheet_id, pwa_uuid	PostgreSQL	One account per business. PWA UUID is unique link for team access.
Item (SKU)	id, account_id, sku, name, description, category, location, quantity, reorder_point, barcode_number, lot_number, expiration_date	Google Sheets	Google Sheets is canonical source. PostgreSQL caches for fast lookup. Sync on change.
Scan Action	id, item_id, action_type (in/out/transfer), from_location, to_location, user_name, timestamp, note, synced_to_sheets	PostgreSQL	Immutable log. Synced flag tracks Google Sheets write status. Offline queue stores here.
Barcode Image	id, item_id, format (Code128), image_url, created_at	Object Storage (S3)	Generated once per item. PNG/SVG format. URL referenced in Sheets.
Label Template	id, account_id, name, layout_type, dimensions, fields_to_print, created_at	PostgreSQL	User-defined or preset (Avery 5160, etc.). PDF generated on-demand.
Location	id, account_id, name, parent_location_id, created_at	Google Sheets + PostgreSQL	Hierarchical: Warehouse > Aisle 3 > Shelf B. Flattened in Sheets for simplicity.

9. GOOGLE SHEETS INTEGRATION SPEC

9.1 OAuth Scopes & Permissions

Required Scopes:

- <https://www.googleapis.com/auth/spreadsheets> (read/write specific spreadsheet)
- <https://www.googleapis.com/auth/drive.file> (create/access files created by app only)

Explicitly NOT Requested:

- Full Drive access (<https://www.googleapis.com/auth/drive>) — user trust barrier
- Gmail access — unnecessary, privacy risk

User Flow:

1. User clicks "Connect Google Sheets"
2. Google OAuth popup: "Inventory Barcode Generator wants to access: [Selected Spreadsheet]"
3. User selects existing spreadsheet OR creates new from template
4. App receives access_token + refresh_token, stores encrypted
5. All subsequent API calls use stored token, auto-refresh on expiry

9.2 Sheet Structure (Auto-Generated)

Tab 1: Inventory Master (User's Existing Data + Appended Columns)

| A: SKU (user) | B: Product Name (user) | C: Location (user) | D: Quantity (user) | E: Barcode Number (app) | F: Last Scan Date (app) | G: Current Location (app) | H: Last Scanned By (app) | I: Reorder Point (user/app) |

Tab 2: Scan History (Auto-Created by App)

| A: Timestamp | B: SKU | C: Action | D: From Location | E: To Location | F: User | G: Note |

Tab 3: Low Stock Alert (Auto-Created, Conditional Formatting)

| A: SKU | B: Product Name | C: Current Quantity | D: Reorder Point | E: Status (Red/Yellow/Green) | F: Suggested Order Qty |

Tab 4: Movement Report (Auto-Created, Weekly Summary)

| A: Week | B: SKU | C: Opening Qty | D: Received | E: Issued | F: Transfers In | G: Transfers Out | H: Closing Qty | I: Variance |

Implementation Notes:

- Appended columns (E–H) use light purple background to distinguish from user data
- Formulas in Tab 3 auto-calculate from Tab 1 data
- Conditional formatting rules applied via Sheets API on first connect
- User can rename tabs; app tracks by internal ID, not tab name

9.3 Sync Logic & Conflict Resolution

Direction 1: PWA Scan → Google Sheets (Push)

- Trigger: User completes scan action in PWA
- Process: API call to Sheets API (append row to Scan History, update Inventory Master row)
- Retry: 3 attempts with exponential backoff. On failure, queue in PostgreSQL, retry every 60 seconds.
- Conflict: If Sheets row modified between read and write, use last-write-wins with timestamp log.

Direction 2: Google Sheets Edit → App Cache (Pull)

- Trigger: Polling every 30 seconds OR webhook (if Sheets API supports)
- Process: Compare last modified time. If newer than cache, pull updated rows.
- Scope: Only Inventory Master tab (not Scan History — that is append-only from app).
- Conflict: If user edited Quantity in Sheets AND scan action updated Quantity, last-write-wins. Log both values in

Scan History for audit.

Offline Handling:

- PWA stores scan actions in IndexedDB queue
- Background sync triggers when connection restored
- If sync fails after 24 hours, email user with pending actions list
- User can manually trigger "Force Sync" from PWA settings

10. PWA MOBILE SCANNER SPEC

10.1 PWA Architecture

Technology Stack:

- Framework: React/Vue with PWA manifest
- Barcode Scanning: ZXing-js or Dynamsoft Barcode Reader (JavaScript, no native app required)
- Camera Access: getUserMedia API with fallback for older devices
- Offline Storage: IndexedDB (via Dexie.js or idb-keyval)
- Background Sync: Service Worker with Background Sync API (where supported) + periodic sync fallback
- State Management: Redux/Vuex with persistence layer

PWA Manifest:

- name: "Inventory Scanner — [Business Name]"
- short_name: "Scan"
- start_url: "/pwa/[uuid]"
- display: "standalone" (hides browser chrome, feels like native app)
- theme_color: "#6c5ce7"
- icons: 192x192, 512x512 (auto-generated with business initials)

Installation:

- iOS: Safari "Add to Home Screen" prompt (detected, shown after 3rd scan)
- Android: Chrome "Install App" banner (auto-triggered on engagement)
- No App Store submission required — bypasses Apple/Google review process

10.2 Scanning Interface

Screen 1: Scanner (Default View)

- Full-screen camera preview with barcode detection overlay (green rectangle on detected barcode)
- Bottom panel: Last 3 scanned items (swipeable list)
- Top bar: User name (tap to change), Sync status (green dot = synced, orange = pending, red = offline), Battery-friendly mode toggle
- Haptic feedback on successful scan (vibration API, 50ms)
- Sound: Optional beep on scan (toggle in settings)

Screen 2: Item Detail (Post-Scan)

- Large product name and image placeholder
- Current Location (editable dropdown)
- Quantity (editable with +/- buttons)
- Action buttons: Check In (green), Check Out (red), Transfer (blue), Just View (gray)
- Note field: Optional text input (e.g., "Damaged in shipping")
- History: Last 5 scan actions for this item (scrollable)
- Done button: Returns to scanner

Screen 3: Transfer Flow

- From Location: Pre-filled (current)
- To Location: Dropdown of known locations OR type new
- Quantity: Default = 1, editable
- Note: Optional
- Confirm: Large button, haptic feedback on success

Screen 4: Offline Queue

- List of pending scan actions with timestamp
- Each item: Action type, SKU, Location, Time, Sync status
- Pull-to-refresh triggers sync attempt

- Settings: "Auto-sync on WiFi only" toggle (for data-conscious users)

10.3 Performance Optimizations

Optimization	Implementation	User Benefit
Camera warmup	Pre-initialize camera on app open (not on first scan). Cache video stream.	First scan < 1 second, not 3–5 seconds
Barcode cache	Cache decoded barcodes in memory for 5 seconds. If same barcode scanned twice, instant lookup.	Rapid sequential scanning (e.g., counting 50 identical items)
Lazy loading	Load item details only when scanned. Keep scanner UI lightweight (< 100KB initial load).	Fast app open on slow 3G connections
Background sync	Use Background Sync API + periodic sync (every 15 min). Fallback to manual sync button.	User never thinks about syncing — it just happens
Battery optimization	Auto-dim screen after 30 seconds of inactivity. Pause camera when app backgrounded.	Full day of scanning without battery anxiety

11. COMPETITIVE DIFFERENTIATION

11.1 Feature Matrix: Us vs. Key Competitors

Feature	Our Product	inFlow	Sortly	Zoho	Fishbowl
Price (unlimited users)	\$39/mo flat	\$99+/mo + per-user	\$49/mo + \$8/user	\$29+/mo + tiered	\$229+/mo + per-user
Google Sheets native sync	■ Source of truth	■ CSV import only	■ Export only	■ Zoho ecosystem	■ QuickBooks only
Setup time	< 1 hour	1–2 hours	30 min	1–2 hours	3+ weeks
Phone camera scanning	■ PWA, offline	■ Native app	■ Native app	■ Limited	■■ Dedicated scanner
Offline mode	■ 500+ actions	■	■	■	■
Check-in/check-out	■ Full workflow	■■ Basic	■	■■ Basic	■
Multi-location transfers	■ Full audit trail	■■ Limited	■■ Limited	■■ Higher tier	■
Movement history	■ Timestamped	■■ Basic	■	■■ Basic	■
Label printing (any printer)	■ PDF, any printer	■	■	■	■ Label printer req
Per-user fees	■ None	■ \$89/user	■ \$8/user	■ Tiered	■ Tiered

11.2 Unique Selling Propositions (USPs)

- **Google Sheets as Source of Truth:** No data migration. No new system to learn. Your spreadsheet gets superpowers, not replaced.
- **Flat Pricing, Unlimited Users:** A team of 10 pays \$39/month, not \$89–\$400+/month. Scaling your team doesn't scale your software cost.
- **Sub-Hour Setup:** From spreadsheet upload to first scan in under 60 minutes. Not 3 weeks. Not 3 days. One lunch break.
- **Offline-First PWA:** Works in warehouses with no WiFi, construction sites in the desert, medical storage with spotty coverage. Syncs when you're back online.
- **Any Printer, Any Phone:** No \$300 scanner hardware. No \$200 label printer. Use what you already own.

12. SUCCESS METRICS & KPIs

12.1 Product Metrics (North Star: Time Saved)

Metric	Target	Measurement	Owner
Time to first barcode	< 10 minutes	Time from account creation to first downloadable barcode PDF	Product
Time to first scan	< 30 minutes	Time from account creation to first successful PWA scan	Product
Inventory count time reduction	80% reduction	Self-reported time before vs. after (survey at 30 days)	Success
Stockout reduction	70% reduction	Self-reported stockout incidents before vs. after (survey at 30 days)	Success
PWA scan success rate	> 95%	Successful decode / total scan attempts (analytics)	Engineering
Google Sheets sync latency	< 5 seconds	Time from scan action to Sheet update visible (server logs)	Engineering
Offline queue sync success	> 99%	Queued actions successfully synced / total queued actions	Engineering

12.2 Business Metrics

Metric	Target (M6)	Target (M12)	Measurement
Monthly Recurring Revenue (MRR)	\$50,000	\$200,000	Stripe dashboard + internal analytics
Paying customers	1,300	5,000	Active subscriptions (any plan)
Trial-to-paid conversion	25%	30%	Trials started / trials converted within 14 days
Annual plan mix	40%	50%	Annual subscriptions / total subscriptions
Net Revenue Retention	110%	120%	(Starting MRR + Expansion – Churn) / Starting MRR
Customer Acquisition Cost (CAC)	<\$50	<\$40	Total sales + marketing spend / new customers
Lifetime Value (LTV)	>\$300	>\$400	Average months active x ARPU x gross margin
LTV:CAC ratio	> 3:1	> 4:1	LTV / CAC
Churn rate (monthly)	< 5%	< 4%	Cancelled subscriptions / total subscriptions

13. RELEASE ROADMAP

13.1 Phase 1: MVP (Months 1–3) — Core Barcode System

Goal: Prove the three-step setup works for Sarah and Jayesh personas.

Features:

- CSV/XLSX upload with auto-column detection
- Bulk Code 128 barcode generation with custom prefixes
- PDF label printing (standard paper + Avery 5160)
- Google Sheets OAuth integration (append barcode columns)
- Basic PWA scanner (camera scan, item lookup)
- Offline mode (store 50 actions, auto-sync)
- User management (email + password, Google Sign-In)

Success Criteria:

- 100 beta users, 50% complete setup within 1 hour
- 80% of beta users generate barcodes within 24 hours of signup
- PWA scan success rate > 90% on iOS and Android
- Google Sheets sync latency < 10 seconds (online)

13.2 Phase 2: v1.1 (Months 4–6) — Asset Tracking & Workflows

Goal: Add check-in/check-out and movement history for Karthik and Hatem personas.

Features:

- Check-in/check-out workflow with reason/note field
- Location transfer with full audit trail
- Movement history dashboard (filter by date, item, user, location)
- Multi-location support (unlimited locations per account)
- Low-stock alerts (email + in-app + Sheets conditional formatting)
- Offline queue expansion to 500 actions
- Basic reporting dashboard (total SKUs, items by location, low stock count)

Success Criteria:

- 25% of users adopt check-in/check-out within 30 days of feature release
- Multi-location accounts grow to 30% of total
- Trial-to-paid conversion reaches 25%

13.3 Phase 3: v1.2 (Months 7–9) — Vertical Expansion

Goal: Capture Dr. Priya (medical) and expand to India/MENA markets.

Features:

- Expiration date tracking with 30/60/90-day alerts
- Lot/batch number tracking for compliance
- Witness scan workflow (two-person verification for controlled substances)
- Tally ERP integration bridge (India market)
- Multi-language label support (English + Arabic + Hindi)
- Shopify/WooCommerce one-click sync
- Advanced reporting (inventory turnover, stock level trends)

Success Criteria:

- Medical vertical accounts reach 10% of total
- India/MENA markets reach 20% of new signups
- Annual plan conversion reaches 40%

13.4 Phase 4: v1.3 (Months 10–12) — Scale & Ecosystem

Goal: Become the platform, not just the tool.

Features:

- API for third-party integrations (Zapier, Make, custom)
- White-label option for resellers/consultants
- Advanced analytics (predictive reordering, demand forecasting)
- Team permissions and role-based access
- Mobile app native wrappers (iOS/Android) for enterprise clients
- Partner marketplace (label printers, barcode scanners, inventory consultants)

Success Criteria:

- MRR reaches \$200,000
- Net Revenue Retention reaches 120%
- API usage by 10% of accounts
- Partner channel contributes 15% of new revenue

14. RISK ASSESSMENT & MITIGATION

Risk	Likelihood	Impact	Mitigation
Google changes Sheets API pricing or limits	Medium	Critical	Build abstraction layer around Sheets API. Maintain PostgreSQL cache as fallback. Monitor API quota usage. Prepare for paid API tier if needed.
Competitor adds Google Sheets sync	High	High	Deepen integration beyond sync: auto-formulas, conditional formatting, custom templates. Build workflow-specific value (check-in/out, transfers) that Sheets alone cannot replicate.
PWA scanning fails on low-end devices	Medium	High	Test on device matrix: iPhone 8+, Samsung A-series, budget Android. Provide manual entry fallback. Consider native app wrapper for enterprise clients.
User churn due to 'good enough' free alternatives	High	Medium	Lock-in via Google Sheets integration depth (formulas, formatting). Time-to-value must be < 1 hour (free tools take days to set up). Annual plan discounts increase switching cost.
Data privacy concerns (GDPR/CCPA/medical)	Medium	High	No PII storage beyond email and user-provided names. SOC 2 Type II certification by Month 9. HIPAA BAA available for medical tier. Clear privacy policy and data deletion workflow.
Scaling costs exceed revenue at low price point	Medium	High	Serverless architecture (AWS Lambda/GCP Cloud Functions) for variable costs. Cache aggressively. PDF generation async (not real-time). Monitor unit economics monthly. Consider usage-based tiers for > 10,000 SKU accounts.
Key persona (Jayesh) requires features we cannot build	Low	Medium	Maintain strict MVP scope. Use concierge onboarding for complex needs. Document feature requests for roadmap. Partner with consultants for custom implementations.

15. APPENDIX: EMPATHY MAP SUMMARY

15.1 Cross-Persona Emotional Patterns

All five personas share a dominant emotional pattern: **I know the answer exists, but every solution I have tried either breaks my workflow, breaks my budget, or breaks my team's trust.**

The aaha moment for all of them is not a feature — it is **This finally works the way I actually work.**

Universal Triggers:

- Crisis mode (not proactive shopping) drives 80% of initial signups
- Previous system failure creates skepticism that must be overcome with speed, not promises
- Time returned (not features gained) is the primary retention driver

Universal Fears:

- Another failed technology investment (Sarah, Jayesh)
- Staff will not adopt (Karthik, Jayesh)
- Compliance/audit failure (Dr. Priya, Hatem)
- Complexity will overwhelm the business (Sarah, Dr. Priya)
- Cost will scale unpredictably with team growth (all)

15.2 Persona-Specific Aaha Moments

Persona	Aaha Moment	Time to Aaha	Retention Driver
Jayesh Suchde	Store manager scanned first item and found 12 missing units in back room — ■400 recovered instantly	< 1 day	Margin impact visibility, investor-ready reports
Karthik Synerg	Senior operator Ramesh (22 years experience) said scanning is faster than his clipboard	3–5 days	Variance reduction proof for MD, peer adoption
Hatem Al Shahed	Client audit passed by sharing Google Sheets link — 3 clicks, full traceability	< 48 hours	Contract retention, multi-site expansion
Sarah Mitchell	Slept in on Sunday morning — first free Sunday in 3 years	1–2 weeks	Weekend time returned, family life restored
Dr. Priya Sharma	Controlled substance count took 8 minutes with two phone scans, not 45 minutes with logbook	1–2 weeks	Compliance confidence, sleep quality restored